

Food, Water, and Energy



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Introduction

The growing demand for water, energy, and food, propelled by a rapidly growing population, puts a serious strain on the ecosystems, leading to environmental deterioration and biodiversity decline. These degradation processes are largely due to lack of coordination between the water, energy and food sectors as each sector operates in isolation, pursuing its own objectives without sufficient consideration of cross-sectoral impacts. This fragmented approach neglects the fact that water, energy and food are interconnected in a complex nexus. Hence, interventions in one sector often have effects on others with sometimes

unintended results. The understanding of this interdependence is further complicated by the development of scientific research and policies along single-discipline silos without accounting for specific interactions. For example, increasing agricultural productivity may intensify irrigation demands, leading to over-extraction of water resources. Similarly, higher energy demands promote large-scale biofuel production that competes with food crops for both land and water. The increased demand for feed for animal production causes large-scale deforestations and biodiversity decline. Consequently, absence of integrated planning across these sectors accelerates environmental degradation and undermines sustainability of food systems. Addressing these challenges requires breaking down disciplinary and institutional divides and adopting nexus-based approaches that recognize and manage interdependencies between water, food, and energy systems.

The water–energy–food (WEF) nexus addresses these challenges by emphasizing cross-sectoral integration and interdependence as a critical pathway to safeguard resource security amid global challenges of climate change, resource scarcity, and competing demands for water, energy, and food. Specifically, the WEF Nexus aims to improve water use efficiency, maintain water quality, concentrate on renewable resources for energy and provide resilient and sustainable food systems.

Moreover, the nexus approach enables *sovereignty* over WEF domains. In the context of the WEF nexus, sovereignty refers to a society's or state's ability to maintain autonomous, secure, and resilient control over its essential water, energy, and food systems, despite external pressures, interdependencies, or cross-sectoral trade-offs. It emphasizes the capacity to make independent decisions about resource use while recognizing that these systems are deeply interconnected and often influenced by global dynamics. In this entry, generic principles and examples will be discussed, with references to studies focusing on the specific interactions, while also paying attention to the challenges the six interactions pose for M&E, a topic that so far has not been integrated in the WEF-centered literature.

WEF Nexus Interactions

According to the initial definition (Hoff2011), the WEF nexus emphasizes the inherent interrelations between water, energy, and food, and provides the analytical framework to reduce trade-offs, build synergies and improve the integrated governance of resources. The WEF nexus acknowledges the strong interrelations between water, energy and food, and aims to capture and clarify the two-way interactions between these resources. Considering the resource flows throughout supply chains, which encompass production, processing, distribution, consumption and management activities, six resource interactions between three resources can be identified, namely: (1) energy inputs in the water system and impact of the water system on energy production or use; (2) energy inputs into the food system, and impacts of the food system on energy production or use; (3) water inputs into the food system and impacts caused by the food system on water quantity and quality; (4) water inputs into the energy system, and impacts of the energy system on water quantity and quality; (5) food inputs into the energy system, impacts by the energy system on food availability and quality; (6) impacts of the water system on food quality and quantity. Although a large literature is now focusing on

the WEF, only few take all six interactions into account simultaneously, particularly in empirical applications (Li et al. 2025).

Interactions

Water-Energy

Water-energy relations are mainly studied in contributions by hydrologists and engineers, following the physical flows of water in the energy system, with little attention paid to their economic or social drivers. Examples of W-E interaction typically include water used in fuel processing and cooling, hydropower generation, and aquavoltaics, utilizing large water bodies for aquaculture, while spanning the surface with photovoltaic panels for energy generation. Typical problems studied include assessment of potential (economically and technically) for hydropower plants, and the impact of such interventions on downstream locations.

Case I: Ignoring the Nexus

Central Asia and the Aral Sea Basin. The Soviet-era water management system for the rivers Amu Darya and Syr Darya prioritized irrigation downstream while upstream countries released water seasonally for hydropower and energy in winter. The immense energy expenditure on constructing and maintaining vast canal systems and pumping stations to divert water for irrigation fields was the fundamental cause of the sea's desiccation (Hamidov et al. 2024).

Case II: Acknowledging the Nexus

Europe: Germany and Netherlands—Urban

Nexus Projects. Cities like Berlin and Rotterdam integrate water and energy flows using smart infrastructure (district heating, wastewater-to-energy, stormwater reuse) with several positive effects like wastewater treatment plants that produce biogas and reduce fossil fuel use. The urban water runoff is reused for non-drinking purposes, reducing water stress. Greenhouse gas emissions are cut and resource efficiency is improved (Deutschland–Nederland (n.d.)).

Water-Food

Water-food interactions are studied by both agronomists, social scientists, and economists in the context of environmental management and improvements in efficiency of resource use. Typical examples of W-F interactions include irrigation in agriculture, water use in food preparation and food processing, while typical problems include optimization of cropping decisions under water scarcity and the impact of innovations in irrigation efficiency on water use efficiency.

Case I: Ignoring the Nexus

Murray–Darling Basin (Australia): Governance, Overallocation, and Poor Integrated Management. Governance reforms and implementation failed to protect environmental flows and coordinate water-for-food decisions. At the basin-scale trade-offs prevailed between irrigation (food production) and river/ecosystem water needs. Consequently, the basin experienced a long-running ecological decline, fish kills and reduced river health. Furthermore, contested water markets and policy failures undermined both agricultural sustainability and the river system. Recent reviews describe multi-decade failures to halt decline despite large investments (Hart 2016; Peerson et al. 2023).

Case II: Acknowledging the Nexus

Taoyuan City, Taiwan: Shihmen Reservoir and Irrigation Ponds. The study reports of a joint operation of a large-scale reservoir together with many small irrigation ponds. Reservoir operations are optimized across short-term (daily) and long-term scales, balancing water supply, hydropower, and food production. Agricultural and public water shortage rates reduced by 22–23%, while food production increased (up to 47% in a wet year) (Uen et al. 2018).

Integrated Aqua-Agriculture in Desalinated Water Setting. The case study shows how aquaculture (tilapia) is combined with hydroponic/herb cultivation (sweet basil) using desalinated water. The integrated water management—food production nexus results in an efficient use of scarce water sources for both food crops and high-value crops. Furthermore, diversification of

fish and herbs improves overall resource productivity and diet diversity (Sewilam et al. 2022).

Food-Energy

The relation between food and energy is often ignored in WEF studies, although there can be clear impact of food policies on the energy sector. Key examples of F-E interactions include the choice between cultivation of food or biofuel crops, the conversion of food waste to energy, and agrivoltaics, the combination of crop production and energy generation through coverage of the cropped area with photovoltaic elements. Typical problems studied here include the impact of timing of planting decisions on energy consumption, and the possibilities to combine crops and photovoltaic panels to enhance resilience of farmers against climate change.

Case I: Ignoring the Nexus

Bioenergy, Food Security and Poverty Reduction: Trade-Offs and Synergies Along the WEF Security Nexus. This study reviews trade-offs and synergies of bioenergy within the WEF security nexus, with emphasis on developing countries. It explores the links of bioenergy with food security, poverty reduction, environmental sustainability, health, and gender equity. In the recent past, ignoring the link between food and biofuels has significantly contributed to the spikes in food prices in 2008, leading to increased food insecurity. The rise of biofuels implies that energy prices are now impacting food prices in two ways: indirectly through costs of fertilizer and energy needed for cultivation and processing, and *directly*, through the coupling of energy and bioenergy prices (Mirzabaev et al. 2014).

Case II: Acknowledging the Nexus

Food Waste Biogas Cogeneration in São Paulo, Brazil. Organic food waste from a food supply center used in anaerobic digestion to produce biogas which powers part of the center. Results are reduced electricity consumption and carbon emissions savings. Furthermore, fertilizer from waste products gave a good return on investment basically because of lowered operating costs (De Oliveira et al. 2024).

Food-Water

Food-water interactions are one of the most studied interactions in the nexus. These interactions mainly focus on the impact of food production on water use and/or pollution of water resources. Typical examples therefore include eutrophication of water resources through the overuse of fertilizers, pollution of water by pesticides and herbicides, but also the computation of virtual water transfers associated with international trade in food, showing that in many cases, virtual water flows go from areas with scarcity to areas with abundance, defying common laws of economics.

Case I: Ignoring the Nexus

Groundwater Depletion in North India. The case study shows that agricultural policies that promote water-intensive crops (e.g., rice, wheat) combined with cheap or subsidized electricity for pumping result in low incentives for efficiency and a weak regulation of groundwater extraction. Essentially, water use for food production expanded without matching consideration for recharge or long-term sustainability (Mukherji 2022).

Case II: Acknowledging the Nexus

Case Study: Robotic Spot Spraying in Sugarcane Farming. A study demonstrates the effectiveness of precision robotic weed spot-spraying technology in sugarcane farming that reduces herbicide use and improves environmental outcomes. Water quality measurements of irrigation-induced runoff showed a 39% reduction in the mean concentration and a 54% reduction in the mean load of herbicides compared to traditional broadcast spraying methods (Rahimi Azghadi et al. 2025).

Energy-Water

Energy-water (E-W) interactions are studied by engineers, agronomists, and (agricultural and energy) economists, focusing on the use of energy in either providing water or the negative impacts of energy generation on water quality or quantity. Typical examples of E-W interactions include desalinization efforts, particularly in water-scarce areas, use of energy for groundwater pumping and

transport of water for irrigation and use as drinking water in general, and wastewater treatment. However, pollution and temperature increases of water caused by energy generation are also seen as an E-W interaction. Typical problems studied therefore include assessing the impact of desalination plants, the sustainability of current irrigation practices, the impact of changes in cropping patterns, the impact of increases in water temperature on life in rivers and deltas, and increasing energy costs to maintain minimum quality standards for irrigation and drinking water.

Case I: Ignoring the Nexus

Pakistan: Hotspots of Energy Use. In Pakistan Punjab and Sindh regions, irrigated crops are energy-intensive due to water pumping using diesel or electricity. Studies have identified “hotspot” districts where energy use and CO₂ emissions per unit of water are especially high, illustrating that water use, energy consumption, and greenhouse gas emissions are tightly linked. Ignoring one dimension can lead to unsustainable outcomes (Siyal and Gerbens-Leenes 2022).

Case II: Acknowledging the Nexus

Solar-Powered Irrigation in Gujarat, India. In Gujarat, India, a pilot project connected 3500 solar-powered irrigation pumps to the grid, with a 25-year agreement to sell surplus power back to the utility. This initiative not only provides renewable energy but also promotes sustainable groundwater use and offers farmers an additional income source. It exemplifies a practical application of the Water-Energy Nexus by linking water, energy, and food systems (Shah et al. 2016).

Energy-Food

Energy-food (E-F) interactions have traditionally been the area of study of (agricultural) economists and agronomists, emphasizing the direct use of energy by agricultural machinery and the indirect impact of fertilizer use. However, E-F interactions span a wider part of the food system, including transport, storage, and processing of food, in commercial plants, but also by households themselves. Hence, typical examples of E-F interactions include agricultural mechanization,

fertilizer use, food transportation and storage, food processing, while also the competition between biofuels and food can be placed here.

Case I: Ignoring the Nexus

Food Versus Fuel: Diversion of Crops Could Cause More Hunger. This study investigates the impact of diverting food crops to biofuel production on global food prices and food security. It highlights that the diversion of food crops to biofuel production was a significant factor contributing to global food prices increasing by 83% in 2008, disproportionately affecting low-income populations.

Assessment of the WEF Nexus in the Life Cycle of Energy Products. A study from Argentina analyses the trade-offs between producing biodiesel and food output. The study shows that ignoring the food–energy nexus creates trade-offs that undermine sustainability. Expanding energy production, especially biofuels, increases the pressure on land and water resources, causing biodiversity loss, and reduce food availability. This weakens food security, escalates environmental degradation, and heightens competition for resources, particularly in regions already facing scarcity and ecological vulnerability (Kock et al. 2024).

Case II: Acknowledging the Nexus

Microgrid Farm and Irrigation Scheduling in Morocco. An off-grid farm in Marrakesh-Safi region using a microgrid with storage and renewable energy to optimize irrigation timing, across crop growth stages. The hybrid system (renewables + storage + smarter irrigation) reduced overall system costs and improved sustainability by lowering dependency on fossil fuels and improving water and energy efficiency (Agadi et al. 2023).

Water-Food-Energy

Lessons Learnt from Case Studies

The lessons learnt from the interactions between the WEF components are summarized in Table 1. The “WEF” row in the table serves as a synthesis of the cumulative lessons from the pairwise

interactions, rather than deriving from a specific standalone WEF case study.

Guidance for Monitoring and Evaluation (M&E)

Because of the complexity of interactions, and interdependence of the three elements in the WEF nexus, designing appropriate frameworks for M&E is a particular challenge. One major problem is that indicators for the individual elements of the WEF are very different, and that, therefore, a composite indicator must be designed that translates the various units to 1, or at least to only a few scores. As described in Sušnik et al. (2018), attempts have been made to construct indices, capturing a part (environmental sustainability, water scarcity) or the whole (human development index) of the nexus, but implementation has been difficult and discussions center around defining the proper metrics as well as on the weighing of different elements.

Hence, the final part of this contribution introduces some guidelines for a stakeholder-validated M&E framework that assesses the performance of the interactions between WEF components and identifies barriers and enablers for the implementation of the nexus. Critical components include a Theory of Change (ToC), clear performance criteria, Key Performance Indicators (KPIs) and underlying metrics, and the application of a rubric methodology, commonly used in educational assessments.

Theory of Change

A Theory of Change (ToC) should define long-term objectives and then work backwards to identify the steps necessary for their achievement, following the trail from long-term impacts to medium-term outcomes and short-term outputs of interventions. Construction of a ToC should engage local institutions and promote knowledge sharing among custodians and policymakers to address constraints that extend beyond technical solutions.

Criteria for Assessment

For uniform WEF Nexus evaluation, clear criteria need to be defined before the intervention start, and the OECD-Development Assistance

Food, Water, and Energy, Table 1 Lessons learnt from the case studies

	Ignoring	Acknowledging
Water-energy	If over-pumping of ground water is ignored the aquifer drawdown will increase the energy demand per unit water pumped	Linking water-saving irrigation with renewable energy improves food security and reduces vulnerability to changing weather patterns Water-efficient cooling systems in power plants conserve freshwater
Energy-water	Inconsistent energy supply affects water deliveries, and created unsustainable water supply dynamics Hydropower plants disrupt downstream ecosystems, compromising water quality and availability for agriculture, fisheries, and human consumption	Coordinated management of water and energy reduces waste in both systems Using renewable energy to power water pumps cuts fossil fuel use
Water-food	Overexploitation of aquifers for agriculture leads to groundwater depletion, causing water scarcity and salinity resulting in a decline in agricultural yields	Using water more judiciously (e.g., drip irrigation, wastewater reuse) sustains food production while conserving scarce supplies
Food-water	Overuse of agricultural fertilizers leads to eutrophication of rivers, lakes, and coastal waters, degrading ecosystems and water quality Virtual water transfers in global food trade reveal that water often flows from scarce to abundant regions	Precision agriculture minimizes the risk of water pollution by reducing excessive fertilizer and pesticide use, optimizing irrigation to prevent nutrient runoff into rivers, lakes, and groundwater
Food-energy	Replacement of food crops by biofuel crops can have serious repercussions on local food supply	Most food waste can serve as feedstock for biofuel production, offering a sustainable alternative energy source while simultaneously reducing landfill waste and greenhouse gas emissions
Energy-food	Reliance on fossil fuels for agriculture undermines resilience and increases greenhouse gas emissions, which in turn affect food systems via climate change	Combine crops and photovoltaic panels to enhance resilience of farmers against climate change
WEF	Failure to balance water allocation between food, energy and other sectors triggers conflict between farmers, pastoralists and consumers Working in silos leads to unintended but avoidable impacts on other sectors (see specific lessons for the six interactions)	Assessing trade-offs justifies fair access to water, energy and food across communities Combing lessons from the six interactions to design interventions that support all three sectors by design

Committee (DAC) criteria are a good point of departure (OECD 2025) to arrive at six assessment criteria as summarized in Table 2. These criteria are particular useful as they can be used in the design of the intervention as well as in the actual assessment, and force policy makers and evaluators alike to critically look at the ToC in terms of measurability of the stated objectives, and the strength and logic of the proposed linkages between interventions and outputs, outcomes and impact.

Key Performance Indicators

Key Performance Indicators (KPIs) focus on the aspects of performance that are most critical to the current and future success of the intervention or

policy, and are compared against pre-defined target values. KPIs are constructed based on underlying metrics, clearly defined entities for which information can be collected, quantitatively or qualitatively. Selection of KPIs is usually based on literature review and extensive stakeholder consultations (Fatima et al. 2025).

Rubrics

As a main challenge for M&E for the WEF nexus is the large divergence in units, a method can be used that is commonly applied in evaluations of educational performance: the design of a set of rubrics that relate KPIs to a predefined scoring metric in a given number of classes. For qualitative assessments, the situation that most closely

Food, Water, and Energy, Table 2 A summary of the OECD-DAC criteria

Relevance	To what extent were the objectives of a WEF intervention consistent with beneficiaries’ requirements, local, regional and country needs and priorities?
Coherence	To what extent were other interventions and policies support or undermine the intended intervention and vice versa?
Effectiveness	To what extent are the development intervention’s objectives achieved, or are expected to be achieved?
Efficiency	To what extent did the intervention pay off in monetary terms when all benefits and costs are taken into account?
Impact	What has been the impact, positively and negatively, primary and secondary of the intervention, directly or indirectly, intended or unintended?
Sustainability	How likely is it that the benefits will continue after the funding of the intervention or policy is ended?

aligns with the described KPI category is selected and, like the quantitative approach, assigned a class. The composition of the rubrics is an important achievement that shows the joint consensus among stakeholders about the classification criteria. This should facilitate a transparent evaluation and avoid idiosyncratic interpretation in post-evaluation periods. Consensus should be reached on KPIs, their measurement and definition of the class boundaries (“what constitutes a good, mediocre, bad, excellent performance for this KPI?”). The next challenge is to weigh the different KPIs, which calls for a transparent framework where the impact of different weights can be easily shown, to encourage stakeholder discussions on the possible trade-offs of interventions, again aligning with the fundamental property of the WEF.

Conclusion

This entry concentrated on the WEF nexus. Since its introduction, many variations on this nexus have been introduced, including the WEF-E (WEF-Ecosystems) nexus, placing emphasis on

biodiversity, land use and ecological services; WEFN (WEF-Nutrition), adding specific focus on nutritional quality of food; an urban nexus (WEF-Wasteland), tailored to assist policy-makers in urban planning and management of waste; and WEFC (WEF-Climate), where feed-back from emissions are explicitly taken into account, just to name a few. This wide variety shows that it is by now fully acknowledged that only system thinking can lead to the design of policies, behavioral change and interventions that are by design environmentally, socially, economically and culturally sustainable.

However, it is also clear that there is still a wide gap between the understanding of systems and the actual willingness and ability of policy makers to use WEF thinking in policy design. M&E frameworks still need to be developed to guide this transition.

Cross-References

- [Energy: Nexus Thinking and Governance](#)
- [Energy: Water-Energy-Food Nexus](#)
- [Water-Energy-Food-Ecosystems Nexus](#)

Competing Interest Declaration The author(s) has no competing interests to declare that are relevant to the content of this manuscript.

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